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SHORT COMMUNICATION

Vulnerability of structures and infrastructure to wildfires: a perspective into assessment and mitigation strategies

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Abstract
Drawing from historical and recent events, this perspective reports the aftermath of the rise in wildfire frequency and intensity, specifically focusing on the vulnerability of structures (i.e., human settlements and infrastructure) and associated losses. We show that the factors contributing to structure losses are diverse and can span environmental/climate dynamics, consequences of urban planning and existing building codes, and human-induced influence. We also present a few case studies of notable wildfire events to offer valuable insights and lessons on how such fires shaped structures and infrastructure losses. This perspective also shows that wildfire risk can be reduced by emphasizing fire-resistant construction, strategic urban planning, public education, and leveraging technological advancements. Finally, this perspective concludes with an emphasis on proactive mitigation measures to safeguard our built environments and communities against the escalating threat of wildfires.

Keywords Wildfires · WUI · Structures · Fire · Losses

1 Introduction

Wildfires, often termed bush/forest or vegetation fires, have long been recognized as critical ecological disturbances (Ferreira et al. 2020). Wildfire occurrence and behavior are a function of complex interactions between climate change effects, vegetation characteristics, and terrain features. The recent rise in wildfires can be tied to dry conditions, extended high temperatures, trapped emissions, forest land encroachment, and urbanization (Paveglio et al. 2018; Sherry et al. 2019). Climate change effects have not only led to the rise in wildfire occurrences but have also led to an increase in casualties, property and ecological losses (Handmer and Betts 2013; Paton et al. 2014). While natural factors are vital in

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triggering wildfires, anthropogenic (i.e., accidental or intentional human-driven causes) can be equally important. Activities such as deforestation, land-use change for agriculture, and urban development disrupt the natural fire cycle and can enhance fire risks and account for a significant proportion of wildfires ("World atlas of natural disaster risk" 2015).

The twentieth century presents a clear shift in which the frequency and intensity of wildfires have soared (Jaafari et al. 2019). More specifically, prolonged dry spells create conducive fire ignition and spread conditions. Further, with higher evapotranspiration rates and drying of potential fuel sources, the window for fire-prone conditions is expanding. For example, statistics show that wildfires may exceed 200,000 incidents and burn 3.5–4.5 million km^2 annually (Tansey et al. 2004). The United Nations Office for Disaster Risk Reduction (UNODRR) reports that a large percentage of wildfires occurred in southern and elevated regions and were amplified due to drought severity (UNDRR 2020).

On a similar front, the number of large fires and the total fire area continue to increase. For example, the average number of large wildfires in the US grew from 140 to 250 in the 1980s to the early 2000s. The National Interagency Fire Center (NIFC) reported a 17% increase in US wildfires in 2022 and a 223% increase since 1983. Further, the US fire season has stretched by about 84 days during 2003–2012 when compared to 1973–1982, which reflects a positive increase of about three days per year (Westerling 2016). Such a fire season seems to be relatively more intense in Western regions, wherein nearly 26,000 wildfires burned approximately 9.5 million acres, compared to 33,000 fires that burned about 0.7 million acres in Eastern areas in 2020. Further, the number of acres burned in Western regions has increased by more than 700% between 1985 and 2017. This notion of a stretched fire season is consistent with that noted by the UNODRR.

As populations grow and technological advancements enable large-scale infrastructure development, this advancement has led to the widespread of the Wildland-Urban Interface (WUI). The twentieth and twenty-first centuries witnessed an exponential expansion of human infrastructure into the WUI. This expansion led to densely populated areas nestled within high fire-risk zones. Modern urbanization, while adopting technologically advanced methods, often overlooks wildfire defensive strategies such as defensible spaces or utilization of fire-resistant building materials. This is likely to intensify WUI's vulnerabilities to wildfire exposure risks. According to Headwaters Economics (Barret 2023), there were more than 99,500 structures that were destroyed by wildfires between 2005 and 2022 (see Fig. 1). Further, the National Fire Protection Association (NFPA) reports that 46 million homes in 70,000 US communities are at risk of wildfires (Bailey 2013).

Naturally, the current urbanization of WUIs requires an expansion of electrical grid infrastructure into these regions (Barbero et al. 2015). This shift suggests that regions that were not previously vulnerable to wildfires may face severe wildfire damages in the near future. A key message from The Center on Global Energy Policy (CGEP) at Columbia University is "*the power sector can be expected to both cause wildfires and be susceptible to them*." This brings a unique dimension to the problem at hand. The LA Times further echoes the CGEP and reports that powerlines ignited twice as many wildfires as any other cause (Penn 2017). The consequences of wildfires extend beyond the immediate destruction of structures and can have cascading effects on local economies and public infrastructure.

Given the multifaceted impacts of wildfires on structures and broader socio-economic systems, there is a critical need to understand these dynamics. This review aims to briefly review some of the notable wildfires, synthesize the current knowledge on structures and infrastructure losses due to wildfires, identify gaps in understanding, and inform strategies for mitigation and adaptation.

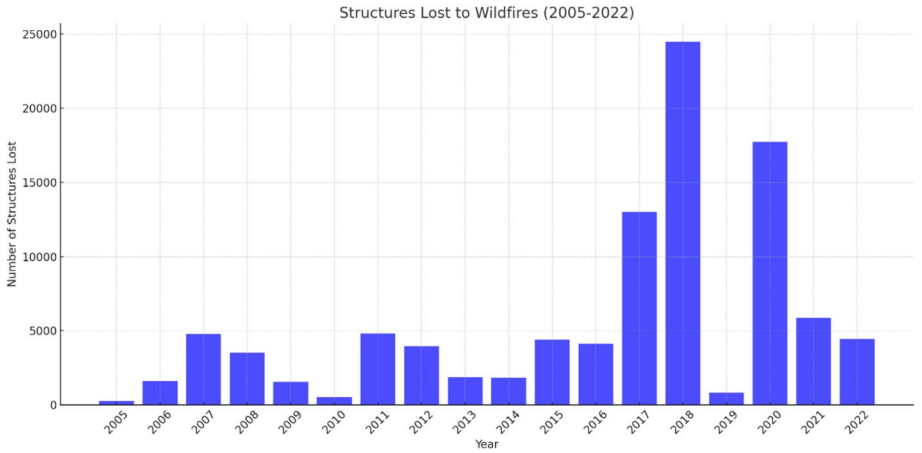


Fig. 1 Number of structures lost to wildfire in the US (Barret 2023)

2 Notable wildfire incidents

To provide more refined pictures of the magnitude of wildfires, a series of notable fires – with a focus on structures and infrastructure losses – is reviewed herein. We start by showcasing historical wildfires and then describe more recent wildfires from around the globe.

2.1 Historical and recent wildfires in the US

According to Roos (Roos 2023), five distinct wildfires are often labeled as the most devastating in the US. These include The Peshtigo Fire and the Great Fires (1871), The Cloquet Fire (1918), The Big Burn (1910), The Camp Fire (2018), The Maui Wildfires (2023), and the Los Angeles Wildfires (2025).

2.1.1 The peshtigo fire and the great fires (1871)

In 1871, four of the most catastrophic fires in US history, collectively designated as the Great Fires of 1871, concurrently impacted the states of Michigan, Wisconsin, and Illinois (Brown 2004). Among these, the Great Chicago Fire resulted in an estimated 300 fatalities, the destruction of approximately 17,500 structures, and left 100,000 individuals without shelter. Simultaneously, the Peshtigo Fire occurred in rural Wisconsin, causing over 2,400 fatalities. Escalating drought conditions drove the initiation of these fires, further worsened by unsustainable logging practices. On October 8, a meteorological front advanced into the Upper Midwest, bringing high-velocity winds that intensified minor ignitions into large-scale wildfires. The Peshtigo Fire alone consumed over 1,200,000 acres of land.

2.1.2 The cloquet fire (1918)

Approximately half a century after the Great Fires of 1871, the Upper Midwest region of the United States was impacted by another significant wildfire event (Cohen 2008). Meteorological conditions analogous to those preceding the 1871 fires occurred, with ignition sources originating from sparks emitted by a traversing locomotive, thereby initiating a wildfire in the vicinity of the small town of Cloquet, Minnesota. Although the fire initially smoldered for several days, it was subsequently reignited by sustained wind velocities of 76 mph. The resulting Cloquet Fire comprehensively consumed the municipalities of Cloquet, Moose Lake, and Kettle River in Minnesota, affecting over 250,000 acres of land. The economic repercussions of this wildfire event are estimated to have incurred damages amounting to $1.42 billion.

2.1.3 The big burn (1910)

The Big Burn is believed to be one of the largest and fastest-growing wildfires in US history. This wildfire converged several wind-driven drought-triggered fires and burned over 3 million acres in Montana, Idaho, and Washington (Blumm 2010). The generated smoke from this fire traveled to New York City and Dallas. Passenger trains were used to evacuate thousands of residents. There were 87 deaths (mainly from firefighters). This wildfire reaffirmed the policy issued by the US Forestry Service to combat and contain all forest fires to avoid conflagrations.

2.1.4 The camp fire in California (2018)

November 2018 witnessed one of California's deadliest and most destructive wildfires. This wildfire originated from a faulty, nearly 100-year-old electrical transmission line in the Sierra Nevada foothills (Spearing and Faust 2020). This fire rapidly grew, given the existing dry conditions and strong winds. The conflagration obliterated nearly 18,800 structures (with 74% being homes). The aftermath damage was estimated at $16.5 billion. The Camp Fire highlighted the need for rigorous infrastructure maintenance and vulnerabilities in evacuation protocols, with arterial roads becoming bottlenecks. Broadly, this wildfire also noted the necessity of updated and modernized building codes and fire-resistant construction materials.

2.1.5 The (Maui) wildfires in Hawaii (2023)

In August 2023, a series of wildfires broke out in Hawaii, predominantly on the island of Maui. These wind-driven fires caused significant damage, with Lahaina being one of the severely affected areas (Mass and Ovens 2024). The fires resulted in at least 115 confirmed fatalities, with others unaccounted for in the aftermath, making it one of the deadliest U.S. wildfires in recent history. Further, the wildfires destroyed over 2,200 structures. The estimated property damage exceeded $6 billion, with significant losses to businesses, cultural landmarks, and infrastructure, including power grids and water systems (Reed 2023).

2.1.6 The Los Angeles wildfires (2025)

As of the writing of this paper, Los Angeles County is tackling a series of devastating wildfires that have imposed significant damage on structures and infrastructure (CALFIRE 2025a). The most prominent among these are the Palisades and Eaton fires. The Palisades Fire ignited in the Pacific Palisades area on January 7, 2025. By January 21, it had burned over 23,713 acres, which resulted in the destruction of approximately 6,528 structures (including the Palisades Branch Library and several educational institutions, including Palisades Charter High School, Cultural landmarks such as the Eames House and the Getty Villa Museum grounds). Concurrently, the Eaton Fire has burned approximately 14,021 acres, including 9,418 structures (CALFIRE 2025b). At the moment, this fire is considered the second most destructive fire in California's history. The fire's impact extended to critical infrastructure, contaminating water supplies managed by the Pasadena Water and Power Department and the Foothill Municipal Water District. Additionally, the Mount Wilson Observatory and several broadcast transmitters were threatened and temporarily disrupted their communication services. As of January 16, 2025, at least 27 fatalities have been confirmed across the affected areas. The scale of the wildfires has necessitated massive evacuation efforts. Over 200,000 residents have been forced to leave their homes. The economic implications of the wildfires suggest that the financial toll could range between $135 billion and $150 billion.

2.2 Wildfires around the world

Here, we describe some notable and recent wildfires that occurred worldwide.

2.2.1 The black dragon fire in China (1987)

Triggered by drought conditions, this wildfire is considered one of the deadliest wildfires in China. The fire destroyed 718 million acres of forest (which is estimated to be 1/6 of China's entire timber reserves). About 211 people died, leaving 50,000 homeless individuals. Not much is reported on the magnitude of structural damage (Zong et al. 2022).

2.2.2 The Muli county fire in Sichuan Province, China (2019 and 2020)

On March 30, 2019, a forest fire erupted in Lier Village within Muli County in Sichuan Province, China. The fire ignited at an altitude of about 3,800 m in an area characterized by complex terrain and limited accessibility. In response, nearly 700 personnel were dispatched to combat the fire. Tragically, a sudden change in wind direction caused an outburst of flames that trapped and killed 27 firefighters and three residents. Then, in late March 2020, Muli County faced another wildfire (Li et al. 2021). On March 28, a fire broke out in a forested area bordering Qiaowa and Xiangjiao townships. Within 15 h, the fire had consumed approximately 30 hectares of forest, which prompted a Grade 3 emergency response.

2.2.3 The Yajiang county fire in Sichuan Province, China (2024)

In March 2024, a significant wildfire ignited near Baizi village in Yajiang County, Sichuan Province, China (He et al. 2024). The fire rapidly spread across approximately 278.8 square

kilometers of mountainous terrain, affecting 250 catchments within the burned area. The region's topography, characterized by high mountains and deeply incised canyons, coupled with strong winds, exacerbated the fire's spread. Over 1,200 firefighters and multiple helicopters were deployed to combat the blaze. The fire necessitated the evacuation of more than 3,000 residents from 11 villages.

2.2.4 The black saturday bushfires in Australia (2009)

The Black Saturday Bushfires took place in February of 2009. These bushfires started by record-breaking temperatures, prolonged drought, and gusty winds across Victoria. The fires destroyed over 2,000 homes and claimed 173 lives, becoming one of Australia's deadliest bushfires (with a total cost close to $4.4 Billion). The Black Saturday Bushfires highlighted the need for rigorous bushfire policies and emphasized the importance of early warning systems and improved evacuation strategies, as well as controlled burns /land management to reduce fuel loads.

2.2.5 The portugal wildfires (2017)

In the summer of 2017, Portugal witnessed a series of large wildfires that broke in the Pedrógão Grande area (Research and 2018 n.d.). These wildfires broke out as a result of high temperatures and thunderstorms. The presence of strong winds enabled the rapid growth and spread of such fires into nearby dense vegetation. Over 500 structures were destroyed, including homes and critical infrastructure. Overall, this wildfire resulted in 66 fatalities and a total cost of EUR 1.5 billion. The tragedy highlighted the importance of community awareness and preparedness, particularly in rural areas.

2.2.6 The canadian wildfires (2023)

In September of 2023, this wildfire burned more than 41 million acres, which is estimated at about 5% of the entire Canadian forest area. Smoke generated from such fires caused air quality alerts and evacuations in Canada and the US, and it crossed Europe. Estimates from structural loss and other types of losses are not yet available at the time of this review.

2.2.7 The Greece wildfires (2023)

The rising average temperature and persistent heatwave that lasted more than two weeks triggered the initiation and spread of the recent Greece wildfires (with other possible causes tied to arson). These fires have caused the death of 28 individuals, evacuation of more than 20,000 people, damage to structures, and cost about half a billion euros ("Wildfires in Greece highlight climate change risks" 2023).

2.3 Other wildfires

Table 1 lists a summary of other wildfires collected from the US and around the world.

Table 1 Summary of notable recent wildfires

Name	Year	Summary	Structures loss
The Texas wildfires (Service 2011)	2011	A collection of severe wildfires that broke due to the 2011 Southern US drought. These fires burned about 47.3% of all burned areas in the US in that year. The resulting economic impact exceeded $3.4 billion. This wildfire resulted in 10 deaths	2,947 homes and over 2,700 other structures
The Miller's Reach Fire (Hollander 2016)	1996	The fire was triggered by drought and low humidity conditions. This wildfire burned over 37,000 acres, causing rail and road transportation between Anchorage and Fairbanks to be temporarily cut off	344 buildings
The Florida firestorm (Minshew and Towle 1999)	1998	This wildfire occurred during the drought and summer of 1998 due to lightning. The blaze crossed natural firebreaks (i.e., rivers and highways) and burned 500,000 acres	150 structures
The Hayman fire (Colorado) ("Hayman fire case study: Summary" 2003)	2002	The cause of this fire was established as arson. This fire burned over 130,000 acres and caused six deaths and the evacuation of 5,400 individuals	133 homes
The Old fire (California) (Keeley et al. 2004)	2003	This wildfire burned 91,000 acres and led to six deaths and about $1.2 billion in damages	975 buildings
The Cedar fire (California) (Brillinger et al. 2009)	2003	This wildfire burned more than 273,246 acres during October and November, costing about $1.3 billion	2,232 residential and 22 commercial buildings
The Tubbs fire (California) (Pausas and Keeley 2021)	2017	Despite not burning as large an area as other fires listed here 36,807 acres, this wildfire remains one of the most destructive in California history. The failure of a private electrical system caused this fire. The Pacific Gas and Electric Company (PG&E) filed for $13.5 billion in bankruptcy	5,643 structures
The Long Draw fire (Oregon) (Robbins 2021)	2012	Triggered by lighting and burned over 550,000 acres. This is considered one of the most significant wildfires in Oregon's history	100s of electric and utility poles

Table 1 (continued)

Name	Year	Summary	Structures loss
The Northern California firestorm (O'Neill et al. 2021)	2017	This wildfire converged several wildfires and burned about 345,000 acres. These fires were considered the costliest at the time, causing around $14.5 billion (2017 USD) in damages, including $11 billion in insured losses, $1.5 billion in fire suppression costs, and 44 deaths	8,900 structures
The Carr fire (California) (Lareau et al. 2018)	2018	This wildfire started due to sparks from a vehicle's tire failure. This fire burned more than 220,000 acres and led to the evacuation of 38,000 individuals. This fire cost $1.6 billion	1,604 structures
The Oregon wildfires (Abatzoglou et al. 2021)	2020	This wildfire burned over 1,200,000 acres and led to the evacuation of 40,000 individuals. Many of these fires were triggered by human action and/or lighting	3,000 structures
The Calf Canyon/Hermits Peak fire (New Mexico) (Bachmeier 2022)	2022	Triggered by dry summer and drought conditions, it burned over 340,000 acres. As its name implies, this wildfire converged into two separate wildfires that were caused by losing control of a prescribed burn and re-ignition of an improperly extinguished pile	900 structures (and threatened over 12,000 others)
The Canadian wildfires (Parisien et al. 2023)	2023	This is a collection of wildfires that broke in 2023, burned 41.92 million acres, and caused the evacuation of over 150,000 individuals. Most fires were initiated via lightning	More than 500 structures
The Portugal wildfires (Research and 2018 n.d.)	2017	The wildfires broke out due to the rising high temperatures and thunderstorms. These fires resulted in 66 fatalities and cost EUR 1.5 billion	About 500 structures
The Black Saturday Bushfires	2009	High temperatures, prolonged drought, and gusty winds triggered these bushfires. These fires destroyed 2,000 homes and cost close to $4.4 Billion	More than 2,000 structures

Table 1 (continued)

Name	Year	Summary	Structures loss
The Maui wildfires (Hawai) (Marris 2023)	2023	Gusting winds from Hurricane Dora, which passed the Hawaiian Islands, blew the wildfire. Evacuation routes were blocked by smoke and heat. The death toll from the Maui wildfires stands at 114, and early cost estimates stand at $6 Billion. The fires burned 17,000 acres	> 2,000 structures
The Camp fire (California)	2018	This wildfire originated from a faulty electrical transmission line. This fire rapidly grew due to the present dry conditions and strong winds. The aftermath damage was estimated at $16.5 billion	> 18,800 structures
The Black Dragon fire (China)	1987	Drought conditions triggered this wildfire. The fire destroyed 718 million acres of forest	–
The Siberian wildfires	2019	This wildfire started in poorly accessible regions of Siberia and burned over 6 million acres	–

3 Structures vulnerable to wildfires

As seen above, wildfires can lead to various structural losses.[1] In general, structures can be grouped under residential/commercial structures, infrastructure (such as bridges and tunnels), utilities (such as electric transmission towers/lines, and communication towers), and cultural and heritage sites. Understanding the nature of these losses can help mitigate future losses and establish new recommendations and guidance.

Residential structures may comprise single- and multi-family homes and apartment complexes. These are typically standalone structures and can be spread across suburban landscapes or nestled within wildland-urban interfaces. Unlike the horizontal and isolated layout of homes, apartment complexes present a different challenge, wherein they house many families and individuals. In addition, their vertical design and proximity of units can lead to rapid horizontal and upward fire spread. The loss of homes imposes economic setbacks and stresses the displacement of families and belongings. In parallel, the consequence of losing an apartment complex can displace multiple families at once, often leading to significant community disruptions, specifically due to the scarcity of alternate housing/accommodations in WUI areas.

To a similar extent, and noting how commercial structures support local economies, the loss of such structures to wildfires could lead to job losses and economic declines. Furthermore, small or minority-owned businesses might not have the financial resilience to recover from such setbacks. In this case, the loss of commercial structures does not directly affect the well-being of their local community but can have a cascading effect on supply chains, especially if the structure is a critical distribution or operational node.

On the other hand, infrastructure and utilities (i.e., bridges, tunnels, electric grid utilities, etc.) serve a much larger region and population than commercial structures. When wildfires attack such structures, then the resulting damage can cripple the mobility of evacuees and first responders. Further, while the electric grid utilities can act as ignition triggers, as seen in many of the reviewed incidents above, the same utilities are also highly vulnerable to fires. In fact, damage to electric lines or substations can lead to widespread power outages, which can impact the population directly affected by the wildfire and other users who may reside away from the fire. This may have a spillover effect, stall emergency responses, hinder information dissemination, and interrupt daily life. In the long term, damaged infrastructure can disrupt essential and logistical services. It is also worth noting that retrofitting or rebuilding infrastructure and utilities often requires significant resources and can take a prolonged period of time.

Cultural and heritage sites are repositories of historical monuments and human achievements. Beyond the immediate architectural and artistic value, these sites also have intangible value in terms of identity, belonging, and shared cultural memory. The loss of such sites to wildfires can demoralize communities and sever ties to historical roots.

[1] From the perspective of this manuscript, "structural loss" can be categorized based on the severity of damage: light damage, where minor repairs are sufficient (e.g., superficial charring, damaged roof tiles); moderate damage, which when occurs can require significant repairs that fall below the threshold for loss in load carrying capacity; and severe damage, where the structure is beyond repair and requires demolition due to loss of functionality. We would also like to point out that other definitions also exist, which may vary depending on the context or criteria used. For instance, definitions can focus on structural integrity, or functional usability to emphasize different aspects of damage.

4 Factors contributing to damage and collapse of structures

According to Barret et al. (2022), structures can ignite from wildfire by one (or a combination) of three means: wind-blown embers, radiant heat, and direct flame contact. The majority of buildings lost in wildfires can be due to embers and surface fires. Since embers can travel long distances, once they land, they can ignite fuel sources in buildings directly by entering through openings or indirectly by igniting nearby combustibles (Hakes et al. 2017). Often, accumulated embers generate enough heat to burn combustibles (vegetation etc.) near buildings or even combustible building materials (Nguyen and Kaye 2021). Radiant heat ignites buildings when nearby materials burn, with a building's susceptibility determined by the intensity and duration of the generated heat. Direct flame contact occurs when flames physically reach a building, leading to its ignition. Overall, the vulnerability of a building or a building component to wildfires (or heat) depends on the intensity and duration of the exposure. Structures can also ignite through less commonly considered mechanisms, such as internal ignition from internal airborne embers, damaged utilities, and unconventional pathways. Internal embers entering through vents, windows, or gaps can ignite furnishings or structural elements. Damaged or poorly maintained gas lines, electrical systems, or utilities can also cause secondary ignition during or after wildfires (Mitchell 2013). Additionally, chemical reactions in stored materials, such as flammable liquids or fertilizers, can serve as ignition sources when exposed to heat or embers. The above pathways also align with those noted by (Manzello 2020).

From the lens of this review, three primary groups of factors can contribute to damage or structure loss in a wildfire. These include environmental factors, urban planning by-laws and building codes, and human-induced factors.

4.1 Environmental factors

The environmental factors inherent to specific landscapes set the foundation for wildfires to ignite and spread (Parisien and Moritz 2009). One such factor is vegetation. Since the moisture content of vegetation determines its flammability, it is natural for dry vegetation to ignite easily. Such dryness can be intrinsic, or it can also be a result of prolonged exposure to sunlight or inadequate water supply. Further, dry vegetation facilitates rapid fire spread due to its ease of combustion. This can be critical in regions with dense layers of vegetation, wherein fires can quickly grow in size and intensity, making them harder to control and extinguish. The dryness of vegetation can also occur due to harsh climate conditions such as heatwaves and droughts.

With everchanging global climate shifts, many regions are experiencing dry spells or unpredictable precipitation patterns, making landscapes more prone to wildfires. The extended heat causes elongated elevated temperatures. Such temperatures lead to increased evapotranspiration rates, which draw more moisture from both soil and vegetation. This can elevate the risk of wildfires even in regions that historically saw fewer fire events (Keeley and Syphard 2021). Additional information on this aspect can be found in the following sources (Guo et al. 2023; Pausas and Keeley 2021; Di Virgilio et al. 2019).

Another environmental factor is wind. At the most basic level, wind provides oxygen for combustion, and when it intensifies, wind can supply increased oxygen levels. Further, the direction and speed of winds significantly influence the spread and growth of wildfires. Winds serve as the primary means to transfer, which can often lead to

igniting spot fires ahead of the primary firefront. These smaller fires can ignite new fires or merge, creating more intense fire events that become increasingly challenging to manage. Wind-driven fires tend to burn more intensely on the windward side (i.e., creating a tilted flame). Such a tilt can cause the fire to spread faster and pre-heat unburnt vegetation (Clements et al. 2008).

4.2 Urban planning and building codes

Urban planning and building codes are crucial components of how structures are designed, constructed, and maintained. As such, these two factors play significant roles in determining the vulnerability of structures during (and post) wildfire events.

The placement of structures in relation to surrounding vegetation is central to predicting how wildfires or embers may reach such development (Nguyen and Kaye 2022a). Naturally, buildings close to dense vegetation are inherently at higher risk as this proximity allows for easier transfer of flames/embers and radiant heat/direct flames to the nearby structures. Proper urban planning can mitigate this risk by imposing buffer space regulations. These regulations prescribe clearing vegetation for a certain distance around structures, creating a buffer zone that can slow or halt the progress of fires. In lieu of reducing the chance of wildfire spread to structures, such buffering provides a safer environment for first responders to combat wildfires (Babrauskas 2020). On the same front, as urban areas expand, the placement of infrastructure that supports them stretches into wildlands and hence becomes critical. For instance, poorly maintained or placed power lines can become ignition sources, as witnessed in some of the reviewed wildfires above. Such infrastructure, unless specifically designed and built with wildfires in mind, can be lost to fire. As such, vegetation clearing practices close to powerlines become critical.

The choice of building materials and architectural designs can determine a structure's survivability during wildfires (Nguyen and Kaye 2022b). Due to its availability and economic viability, regional preferences for specific construction materials, such as wood, can take precedence over other non-combustible materials. Advancements in building sciences have introduced fire-resistant materials, including non-combustible siding, tempered glass, and metal roofing. These materials can be integrated into modern construction through innovations such as modular designs, advanced coatings, and composite systems that enhance durability and thermal resistance. For example, intumescent coatings that expand under heat to form protective barriers can delay the spread of flames and reduce structural damage. Additionally, the use of porous and lightweight aerogels can provide improved insulation while maintaining fire resistance. Fire-resistant engineered wood, treated with chemicals like borates, improves fire performance and maintains structural integrity under high temperatures. Advanced composite systems can offer enhanced load-bearing capacity and resistance to thermal degradation.

Incorporating these materials can enhance a structure's resistance to fires. Moreover, designs that minimize openings and gaps can further reduce the risk of ignition. Eaves and decks can trap embers and turn into channels that direct fire toward the main structure. As such, these elements need to be properly detailed. Specific building code provisions, with mitigation strategies for overcoming wildfire hazards, can mandate the use of such construction materials. Areas prone to wildfires can greatly benefit by enforcing such codes.

4.3 Human-induced factors

Whether intentional or accidental, there is a large human factor in wildfire breakout and control (Aldersley et al. 2011). For example, recreational activities (camping, etc.) have often been linked to wildfire initiation. Such wildfires may start from barbecues and smoking/celebrations. While they often start small, poorly maintained fires can grow in wildfires. Similarly, machine or vehicle sparks can ignite dry grass/leaves. As mentioned above, poorly maintained electric infrastructure can come into contact with vegetation and cause sparks. Evidently, arson and vandalism are concerning human-induced factors, as seen in recent wildfires in Greece.

On a different front, the lack of land management practices can amplify the risk of wildfires. For instance, despite its good intentions, policies focused on suppressing all fires might have inadvertently increased wildfire risks by preventing smaller and controlled fires to reduce dense vegetation accumulation naturally. If land clearing is not followed properly, it can expose the soil to direct sunlight, decreasing its moisture content and making it more susceptible to fires. Similarly, if prescribed fires are not extinguished properly, then these can reignite and grow into wildfires (Penman et al. 2014).

5 Economic and social impacts

As Table 1 has demonstrated, wildfires can have major economic and social consequences. The impacts of wildfires go beyond the immediate, direct damages and may spread through various parts of the affected communities through indirect losses and adverse social impacts.

Over 46 million homes in 70,000 urban, suburban, and native communities are at risk of wildfires in the US alone (Bailey 2013). For example, losses due to wildfires continue to grow annually. The past years have also observed related incidences, most notably in Greece, Portugal, Canada, and, most recently, in Australia. One should also be cognizant that in a severe fire season, wildfires can destroy thousands of structures (e.g., 10,488 buildings in the 2020 California wildfires (CALFIRE 2020) and 5,900 buildings in the 2020 Australian bushfires (Tin et al. 2020)). While the average insurance loss due to wildfires in the US is approximately $3.5 billion (as noted in the last twenty years), the overall losses may exceed $20 billion per wildfire incident (Barret 2021).

5.1 Direct economic losses

Direct economic losses provide a visible metric immediately apparent after a wildfire, as it describes the cost of the immediate destruction caused by a wildfire. This metric translates into the market value of lost structures and includes any potential earnings or services they might have generated. A portion of this metric is insurance claims, which often surge after a wildfire. Such a sudden influx of claims can strain insurance companies. This can also lead to imposing higher premiums (or withdrawing coverage) for structures located in fire-prone areas (Handmer and Betts 2013).

5.2 Indirect economic losses

While direct losses offer visible metrics, indirect economic losses are often dispersed over time and space. For example, wildfires can lead to prolonged business disruptions, specifically when structures and critical infrastructure are severely damaged or destroyed. Beyond the direct damage to commercial properties, the loss of power, disrupted supply chains, and impeded transportation routes can stall business operations. Small businesses are naturally vulnerable to such disruptions and can struggle to resume operations. This can also result in job losses and decreased local economic activity. Further, wildfire-vulnerable regions might witness declining property values – especially if insurance premiums are high or withdrawn. The perceived risks of wildfires can deter economic or federal investments. Such an action can have significant social implications (Calkin et al. 2011).

5.3 Social impacts

Other than monetary metrics, wildfires have a number of social implications. An immediate consequence of a wildfire is the displacement of residents by means of evacuations. Such evacuations can affect thousands of families and could become logistical challenges. The displaced residents, especially if their properties were destroyed, are taxed with the stressful task of rebuilding their lives. In addition, evacuees and survivors often undergo the trauma of experiencing a wildfire and the destruction of their homes, schools, or businesses. Such trauma can have lasting psychological impacts (Finlay et al. 2012). Moreover, the smoke and pollutants from wildfires can exacerbate respiratory conditions, posing health risks to vulnerable populations.

6 Mitigation and prevention strategies

With the continued rise in consequences of wildfires, there is an emphasis on mitigation and prevention strategies. Some of such strategies are briefly described herein. We invite interested readers to visit the cited sources and others to gain further insight and details. In many instances, a large body of literature exists on lessons learned from recent wildfires that can be adopted to help fight future events.

6.1 Building codes and practices

Many modern construction materials, such as new types of concrete, steel, engineered wood, FRP, exhibit lower fire performance than traditional construction materials. Incorporating fire-resistant construction materials can be a vital strategy for minimizing wildfire damages as these materials can better withstand high temperatures and slow down the spread of fire to allow first responders to respond to the fire and occupants to evacuate. For example, in one study, Barret et al. (Barrett et al. 2022) noted that 94.2% of the structures analyzed by their team (in CA wildfires) have wildfire-resistant and non-combustible roof coverings. However, the rest (5.8% of total assessed buildings) had wood roofs vulnerable to wildfires. Barret et al. estimate retrofitting these roofs would cost a minimum of $6 billion (Barret 2022).

The practice of using defensible spaces where vegetation, debris, and other potential fire fuels are cleared/reduced can help improve the fire safety of structures. For example,

defensible spaces act as a buffer that can slow or stop the spread of flames to structures. The National Fire Protection Association presents a systematic approach to segregating such a space into zones, each with a specific purpose. For instance, the closest zone to a structure might be empty of vegetation, while subsequent zones may have reduced vegetation.

Currently, the design and construction of infrastructure utilities, such as transmission towers, do not take into account the potential hazards of wildfires. It is worth noting that the main construction guiding documents for such structures do not contain guidelines or mention wildfires (ASCE 2015; Wong and Miller 2009). Similarly, at the moment, only a few states (i.e., California, Nevada, Pennsylvania, and Utah) in the US have a statewide building code requiring wildfire-resistant measures for new construction (Barrett et al. 2022).

6.2 Incorporating structural resiliency

Designing critical structures and infrastructures for fire resiliency is a natural response to the increasing wildfire threats. This imperative stems from recognizing that structural integrity serves as the final line of defense when other fire protection measures, such as sprinklers and firefighting efforts, fail (Baker et al. 2020). In parallel, ensuring adequate fire resiliency in structural systems is essential for minimizing injuries and fatalities among occupants and firefighters and preventing structural collapse, which further minimizes rehabilitation costs.

Currently, building codes mandate infrastructures to be designed to withstand extreme loading events, including earthquakes, blasts, and hurricanes. However, there is a notable deficiency in specific requirements or provisions for designing structures with fire resiliency in mind. This gap is particularly evident in the absence of fire resistance standards for bridges and tunnels, which are vital components of transportation networks. The lack of fire-resilient design criteria for such infrastructures challenges their ability to withstand and recover from wildfire impacts, potentially leading to prolonged disruptions and increased vulnerability of communities. The same can also be true during wildfires, as such infrastructure is commonly used for evacuation and firefighting activities.

Implementing fire-resilient design practices involves using fire-resistant materials, enhanced structural detailing, and incorporating passive fire protection systems. For instance, employing these strategies can significantly reduce the spread of fire and maintain structural integrity during wildfires. Designing bridges and other critical infrastructures with redundancy and robustness can ensure their continued functionality even under extreme fire conditions. Therefore, it can be quite of merit to integrate these measures into building codes and standards as a means to improve the resilience of structures (Carton et al. 2024).

6.3 Urban planning and zoning regulations

Another means that has the potential to serve as a mitigation strategy is to follow conscious urban planning that integrates fire safety considerations. This approach currently exists, with varying degrees, and generally involves identifying and zoning high-risk areas susceptible to wildfire hazards (Mockrin et al. 2020). This zoning process relies on comprehensive analyses of historical fire data, vegetation types, topographical features, climate conditions, etc. When these factors are taken into consideration, urban planners can strategically limit or prohibit development in areas with elevated wildfire risks. In regions where

development is permitted, thought-in-process shifts to enforce rigorous regulations to mandate the construction of structures with high levels of fire resiliency. This includes the use of fire-resistant materials, the incorporation of defensible space around buildings, to name a few.

Furthermore, urban planning in wildfire-prone areas can incorporate infrastructural enhancements to facilitate efficient evacuation and firefighting operations (Gonzalez-Mathiesen and March 2018). Such enhancements may include wider roads, which can be advantageous for enabling the rapid movement of emergency vehicles and unobstructed evacuation routes during evacuations. Further, the inclusion of refuge areas can also provide safe zones for individuals awaiting. Additionally, the establishment of centralized and independent water sources can support firefighting efforts by ensuring a reliable and accessible supply of water for fire suppression activities.

6.4 Public awareness and education

Public awareness and education play a fundamental role in enhancing community resilience against wildfires (Rodrigo-Comino and Salvati 2024). For example, effective dissemination of information and comprehensive education programs can be helpful to ensure that residents are well-prepared to respond appropriately during wildfire incidents, which in turn can reduce potential casualties and property loss. Hence, engaging residents through workshops, informational campaigns, and interactive sessions fosters a culture of preparedness and mutual support. This is due to the notion that when communities are actively involved in planning and decision-making processes related to wildfire management, there is a higher level of cooperation among residents and local authorities. Thus, regular drills and training sessions can be integral components of this preparedness strategy. According to (Prestemon et al. 2010), systematic evacuation drills equip individuals with the necessary skills and knowledge to execute evacuation procedures efficiently. These exercises not only familiarize residents with the logistical aspects of evacuation but also reduce confusion during wildfire events.

Clearly and strategically placing signage is another element that can facilitate efficient evacuations. Signages for evacuation routes, assembly points, and safety zones can allow residents to navigate away from danger effectively – especially in high-stress situations where the cognitive load is increased, and quick decision-making is required. It is also equally important for these signs to be well-maintained and easily interpretable.

6.5 Technological advancements

The evolution of firefighting technologies for wildfires has undergone significant transformations over time (De et al. 2019). Early firefighting initially relied primarily on ground activities via manual labor and tools such as axes, etc. These methods were labor-intensive and often ineffective against large-scale wildfires, as seen in some notable wildfire incidents reviewed herein. Then, in the mid-twentieth century, the introduction of aerial firefighting marked a key advancement. This technology enabled firefighters to access remote and inaccessible areas rapidly. This capability not only enhanced the efficiency of suppression efforts but also allowed for the strategic application of fire retardants to create firebreaks and slow the spread of wildfires. The latter part of the twentieth century and the early twenty-first century saw significant technological innovations aimed at improving both the prevention and suppression of wildfires. The development of Geographic

Information Systems (GIS) and advanced fire modeling software allowed for more accurate fire behavior predictions, which helps enable better resource allocation and preemptive measures.

More recently, automation and artificial intelligence have further revolutionized wildfire firefighting technologies. Nowadays, autonomous drones and other sensors can monitor large areas continuously to identify hotspots and assess the effectiveness of suppression efforts without direct human intervention. Additionally, advancements in robotics have led to the development of autonomous firefighting vehicles capable of navigating challenging terrains and delivering water or fire retardants with precision. Similarly, the integration of the Internet of Things (IoT) allowed the deployment of advanced sensors at strategic locations in forests and to collect real-time data on environmental conditions. These data streams are processed and analyzed to provide actionable insights and proactive measures to mitigate wildfire risks. Furthermore, there is much more focus on using eco-friendly fire retardants and energy-efficient firefighting equipment to minimize the ecological footprint of suppression activities. Thus, mitigation strategies may leverage modern technologies to enhance our ability to predict, combat, and recover from wildfires. For example, satellite imagery, ground-based sensors, predictive algorithms, and early warning systems can provide real-time data on potential fire outbreaks (Naser and Kodur 2020).

7 Conclusions

This review of the increasing frequency and intensity of wildfires highlights a central aspect of resilience in the face of evolving infrastructure-related challenges. The review also advocates for the adoption of fire-resistant construction methods, strategic urban planning, and public education to mitigate wildfire risks. These recommendations are vital for enhancing the resilience of human settlements and infrastructure against wildfires. Furthermore, the historical and recent data presented in the review illustrate the escalating threat of wildfires due to climate change effects, such as extended droughts, high temperatures, and changes in vegetation and terrain. In reviewing the complex dimensions of wildfires and their impacts on structures, the following key points emerge:

- The historical frequency and intensity of wildfires continue to increase as a result of a complex interplay between natural climatic shifts and anthropogenic factors.
- With the expansion of urbanization into wildfire-prone areas (WUI), the risks to structures and infrastructures have significantly grown.
- The aftermath of wildfires is no longer confined to immediate structures but can now comprise direct and indirect economic implications with a distinct social impact.
- In the wake of devastating wildfires, the emphasis is on realizing practical and proactive measures and mitigation strategies.
- Ensuring fire resiliency in structures can be a highly effective measure for mitigating damage to structures under wildfires.
- There is a need to further examine the wildfire vulnerability of rural homes, villas, industrial facilities, and other types of buildings, as such structures and industrial facilities are at greater risk from wildfires.

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Data availability Some or all data, models, or code that support the findings of this study are available from the corresponding author upon reasonable request.

Declarations

Conflict of interest The authors declare no conflict of interest.

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References

Abatzoglou JT, Rupp DE, O'Neill LW, Sadegh M (2021) Compound extremes drive the western oregon wildfires of september 2020. Geophys Res Lett. https://doi.org/10.1029/2021GL092520

Aldersley A, Murray SJ, Cornell SE (2011) Global and regional analysis of climate and human drivers of wildfire. Sci Total Environ. https://doi.org/10.1016/j.scitotenv.2011.05.032

ASCE (2015) Design of Latticed Steel Transmission Structures. ANSI/ASCE Standard, American Society of Civil Engineers, (10 97), 1–69

Babrauskas V (2020) Firebrands and Embers. Encyclopedia of Wildfires and Wildland-Urban Interface (WUI) Fires

Bachmeier S (2022) Calf Canyon Fire produces a pyrocumulonimbus cloud — CIMSS Satellite Blog, CIMSS. CIMSS Satellite Blog, <https://cimss.ssec.wisc.edu/satellite-blog/archives/46218> (Sep. 10, 2023)

Bailey D (2013) WUI Fact Sheet

Baker G, Webb A, Whiting P (2020) Regulatory controls for buildings in wildfire-prone areas of Australia. Fire Technol. https://doi.org/10.1007/s10694-020-00999-7

Barbero R, Abatzoglou JT, Larkin NK, Kolden CA, Stocks B (2015) Climate change presents increased potential for very large fires in the contiguous United States. Int J Wildland Fire. https://doi.org/10.1071/WF15083

Barret K (2021) Wildland-Urban Interface: The Problem, Trends, & Solutions (slides) - Headwaters Economics. Headwater Economics, <https://headwaterseconomics.org/wildfire/homes-risk/wui-problem-trends-solutions/> (Sep. 9, 2023)

Barret K (2022) Wood roofs are a $6 billion wildfire problem. Headwaters Economics, <https://headwaterseconomics.org/natural-hazards/wood-roofs-wildfire/#methods> (Sep. 9, 2023)

Barret K (2023) Wildfires destroy thousands of structures each year

Barrett K, Quarles SL, Gorham DJ (2022) Construction Costs for a Wildfire-Resistant Home California Edition Author Information Contact

Blumm MC (2010) Present at the Creation: The 1910 Big Burn and the Formative Days of the US Forest Service. Ecology Law Quarterly

Brillinger DR, Autrey BS, Cattaneo MD (2009) Probabilistic risk modeling at the wildland urban interface: the 2003 cedar fire. Environmetrics. https://doi.org/10.1002/env.959

Brown H (2004) 'The air was fire': fire behavior at Peshtigo in 1871. Fire Management Today

CALFIRE (2020) 2020 Fire Season - CALFIRE. <https://www.fire.ca.gov/incidents/2020/> (Nov. 16, 2020)

CALFIRE (2025a) Palisades Fire: Incident Update on 01/21/2025 at 6:32 PM | CAL FIRE. <https://www.fire.ca.gov/incidents/2025/1/7/palisades-fire/updates/6cb6ffe9-3b14-47fd-849a-b3debb163afd> (Jan. 22, 2025)

CALFIRE (2025b) Eaton Fire | CAL FIRE. <https://www.fire.ca.gov/incidents/2025/1/7/eaton-fire> (Jan. 22, 2025)

Calkin DC, Finney MA, Ager AA, Thompson MP, Gebert KM (2011) Progress towards and barriers to implementation of a risk framework for US federal wildland fire policy and decision making. For Policy Econ. https://doi.org/10.1016/j.forpol.2011.02.007
Carton H, Gales J, Kennedy EB (2024) Identifying Research Needs for Canadian Wildfire Building Code Development. Lecture Notes in Civil Engineering, Springer, Cham, 495 LNCE, 15–27
Clements CB, Zhong S, Bian X, Heidman WE, Byun DW (2008) First observations of turbulence generated by grass fires. J Geophys Res Atmos. https://doi.org/10.1029/2008JD010014
Cohen J (2008) The wildland- urban interface fire problem. Forest History Today
De DK, Olawole OC, Joel ES, Ikono UI, Oyedepo SO, Olawole OF, Obaseki O, Oduniyi I, Omeje M, Ayoola AA, Usikalu MR, Omotosho TV, Bolujo EO, Ilo IP (2019) Twenty-first century technology of combating wildfire. IOP Conference Series: Earth and Environmental Science.
Ferreira LN, Vega-Oliveros DA, Zhao L, Cardoso MF, Macau EEN (2020) Global fire season severity analysis and forecasting. Comput Geosci. https://doi.org/10.1016/j.cageo.2019.104339
Finlay SE, Moffat A, Gazzard R, Baker D, Murray V (2012) Health impacts of wildfires. PLoS Currents. https://doi.org/10.1371/4f959951cce2c
Gonzalez-Mathiesen C, March A (2018) Establishing design principles for wildfire resilient urban planning. Plan Pract Res 33(2):97–119
Guo D, Saft M, Hou X, Webb JA, Hairsine PB, Western AW (2023) How does wildfire and climate variability affect streamflow in forested catchments? a regional study in eastern Australia. J Hydrol 625:129979
Hakes RSP, Caton SE, Gorham DJ, Gollner MJ (2017) A Review of pathways for building fire spread in the wildland urban interface part II: response of components and systems and mitigation strategies in the United States. Fire Technol 53(2):475–515
Handmer J, Betts R (2013) Estimating the economic, social and environmental impacts of wildfires in Australia Catherine Stephenson1. Environmental Hazards
Hayman fire case study: summary. (2003). USDA Forest Service - General Technical Report RMRS-GTR
He K, Hu X, Wu Z, Zhong Y, Zhou Y, Gong X, Luo G (2024) Preliminary analysis of the wildfire on March 15, 2024, and the following post-fire debris flows in Yajiang County, Sichuan, China. Landslides, Springer Sci Bus Media DtschLand GmbH 21(12):3179–3189
Hollander Z (2016) Big Lake marks 20th anniversary of Millers Reach, the wildfire that changed everything—Anchorage Daily News. ADN Logo
Jaafari A, Zenner EK, Panahi M, Shahabi H (2019) Hybrid artificial intelligence models based on a neuro-fuzzy system and metaheuristic optimization algorithms for spatial prediction of wildfire probability. Agric for Meteorol 266–267:198–207
Keeley JE, Fotheringham CJ, Moritz MA (2004) Lessons from the October 2003 wildfires in Southern California. J for 102(7):26–31
Keeley JE, Syphard AD (2021) Large California wildfires: 2020 fires in historical context. Fire Ecol. https://doi.org/10.1186/s42408-021-00110-7
Lareau NP, Nauslar NJ, Abatzoglou JT (2018) The carr fire vortex: a case of pyrotornadogenesis? Geophys Res Lett. https://doi.org/10.1029/2018GL080667
Li Q, Cui J, Jiang W, Jiao Q, Gong L, Zhang J, Shen X (2021) Monitoring of the fire in Muli County on March 28, 2020, based on high temporal-spatial resolution remote sensing techniques. Nat Hazards Res 1(1):20–31
Manzello SL (2020) Structure ignition mechanisms in wildland-urban interface (WUI) fire and associated california state fire marshal standard test methods
Marris E (2023) Hawaii wildfires: did scientists expect Maui to burn? Nature NLM (Medline) 620(7975):708–709
Mass C, Ovens D (2024) The Meteorology of the August 2023 Maui Wildfire. Weather Forecast, Am Meteorol Soc 39(8):1097–1115
Minshew P, Towle J (1999) The 1998 wildfires in central Florida: Volusia County's own Armageddon. J Environ Health 61(7):22
Mitchell JW (2013) Power line failures and catastrophic wildfires under extreme weather conditions. Eng Fail Anal 35:726–735
Mockrin MH, Fishler HK, Stewart SI (2020) After the fire: perceptions of land use planning to reduce wildfire risk in eight communities across the United States. Int J Disaster Risk Reduct 45:101444
Naser MZ, Kodur VKR (2020) Concepts and applications for integrating unmanned aerial vehicles (UAV's) in disaster management. Adv Comput des 5(1):91–109
Nguyen D, Kaye NB (2021) Experimental investigation of rooftop hotspots during wildfire ember storms. Fire Saf J 125:103445

Nguyen D, Kaye NB (2022a) The role of surrounding buildings on the accumulation of embers on rooftops during an ember storm. Fire Saf J 131:103624
Nguyen D, Kaye NB (2022b) Quantification of ember accumulation on the rooftops of isolated buildings in an ember storm. Fire Saf J 128:103525
O'Neill SM, Diao M, Raffuse S, Al-Hamdan M, Barik M, Jia Y, Reid S, Zou Y, Tong D, West JJ, Wilkins J, Marsha A, Freedman F, Vargo J, Larkin NK, Alvarado E, Loesche P (2021) A multi-analysis approach for estimating regional health impacts from the 2017 Northern California wildfires. J Air Waste Manag Assoc 71(7):791–814
Parisien MA, Barber QE, Bourbonnais ML, Daniels LD, Flannigan MD, Gray RW, Hoffman KM, Jain P, Stephens SL, Taylor SW, Whitman E (2023) Abrupt, climate-induced increase in wildfires in British Columbia since the mid-2000s. Commun Earth Environ 4(1):1–11
Parisien MA, Moritz MA (2009) Environmental controls on the distribution of wildfire at multiple spatial scales. Ecol Monogr 79(1):127–154
Paton D, Buergelt PT, McCaffrey S, Tedim F (2014) Wildfire hazards, risks, and disasters
Pausas JG, Keeley JE (2021) Wildfires and global change. Front Ecol Environ 19(7):387–395
Paveglio TB, Edgeley CM, Stasiewicz AM (2018) Assessing influences on social vulnerability to wildfire using surveys, spatial data and wildfire simulations. J Environ Manage 213:425–439
Penman TD, Bradstock RA, Price OF (2014) Reducing wildfire risk to urban developments: simulation of cost-effective fuel treatment solutions in south eastern Australia. Environ Modell Soft 52:166–175
Penn I (2017) Power lines and electrical equipment are a leading cause of California wildfires - Los Angeles Times. LA Times, <https://www.latimes.com/business/la-fi-utility-wildfires-20171017-story.html> (May 2, 2022)
Prestemon JP, Butry DT, Abt KL, Sutphen R (2010) Net benefits of wildfire prevention education efforts. For Sci 56(2):181–192
Reed B (2023) Bare power lines and 'obsolete' poles were possible cause of Hawaii fires | Hawaii fires | The Guardian. *Guardian US*, <https://www.theguardian.com/us-news/2023/aug/26/hawaii-wildfires-possible-cause-bare-power-lines> (Sep. 10, 2023)
Research DV-F, 2018, undefined. (n.d.). Wildfires in Portugal. *scholar.archive.org*
Robbins WG (2021) Oregon and climate change: the age of megafires in the American West. Or Hist Q 122(3):250
Rodrigo-Comino J, Salvati L (2024) Fire hazards: socio-economic and regional issues. Springer, Cham
Roos D (2023) The 5 Deadliest Wildfires in America | HISTORY. History, <https://www.history.com/news/deadliest-wildfires-north-america> (Sep. 9, 2023)
Service TAF (2011) 2011 Texas Wildfires
Sherry J, Neale T, McGee TK, Sharpe M (2019) Rethinking the maps: a case study of knowledge incorporation in Canadian wildfire risk management and planning. J Environ Manage 234:494–502
Spearing LA, Faust KM (2020) Cascading system impacts of the 2018 camp fire in California: the interdependent provision of infrastructure services to displaced populations. Int J Disaster Risk Reduct 50:101822
Tansey K, Grégoire JM, Stroppiana D, Sousa A, Silva J, Pereira JMC, Boschetti L, Maggi M, Brivio PA, Fraser R, Flasse S, Ershov D, Binaghi E, Graetz D, Peduzzi P (2004) Vegetation burning in the year 2000: global burned area estimates from SPOT VEGETATION data. J Geophys Res d: Atmos. https://doi.org/10.1029/2003JD003598
Tin D, Hertelendy AJ, Ciottone GR (2020) What we learned from the 2019–2020 Australian Bushfire disaster: making counter-terrorism medicine a strategic preparedness priority. Am J Emerg Med 46:742
UNDRR (2020) Disaster risk reduction in India: status report 2020. United Nations Office for Disaster Risk Reduction (UNDRR), Regional Office for Asia and the Pacific
Di Virgilio G, Evans JP, Blake SAP, Armstrong M, Dowdy AJ, Sharples J, McRae R (2019) Climate change increases the potential for extreme wildfires. Geophys Res Lett 46(14):8517–8526
Westerling ALR (2016) Increasing western US forest wildfire activity: sensitivity to changes in the timing of spring. Philos Trans R Soc b: Biol Sci 371(1696):20150178
Wildfires in Greece highlight climate change risks. (2023). *Emerald Expert Briefings*, Oxford Analytica, oxan-es(oxan-es)
Wong CJ, Miller MD (2009) Guidelines for electrical transmission line structural loading. American Society of Civil Engineers, Reston
World atlas of natural disaster risk. (2015). Choice reviews online.
Zong X, Tian X, Yao Q, Brown PM (2022) An analysis of fatalities from forest fires in China, 1951–2018. Int J Wildland Fire 31(5):507–517

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